A Two-Frequency Sub-Millimetre Dish Search at 542 and 596 GHz: A Pre-Registered Kill Test for a Dark-Energy-Tied Gravity Framework

Working note, 2026-10-15, v1.1 (corrected: decisive coupling is $\sim 10^{-13}$, see §6). AI-assisted research programme; not peer reviewed.

Abstract

We propose a narrow-band dish-antenna search for a 0.5 mm line at two fixed frequencies, 541.7 GHz and 595.5 GHz.

The framework and its prediction. We test a framework in which the galactic acceleration scale a_0 is tied to the dark-energy density, $\rho_{\rm DE} = 4a_0^2/(Gc^2)$. In this framework, the extra gravitating component ("cold energy") has no non-gravitational channel. The only quantum mass the framework's vacuum induces is $m = (\rho_{\rm DE}\hbar^3/c^3)^{1/4}$. That gives 2.240 meV for the canonical value of a_0 and 2.463 meV for the alternative value.

What the search decides. If the cold component did couple to light as a kinetically mixed vector field of this mass, a metal plate would radiate a line at exactly these frequencies.

- **A detection would falsify the framework.** The framework predicts no such line, and this prediction was pre-registered.
- **A null result would close** the only portal through which the cold component could be converted to light at useful efficiency.

Reach. A $1\,\mathrm{m}^2$ plate with a warm receiver reaches a mixing $\varepsilon \approx 7 \times 10^{-12}$ in one hour. A $10\,\mathrm{m}^2$ plate with an SIS receiver reaches 1.4×10^{-13} in 30 days. **Correction (v1.1):** CMB spectral-distortion bounds already exclude $\varepsilon \gtrsim 10^{-13}$ at this mass for a vector field present since the early universe (provisional), so the decisive search must reach $\varepsilon \sim 10^{-13}$: the $10\,\mathrm{m}^2$, 30-day configuration. No published search covers 1–3 meV; the nearest reached $\chi \lesssim 10^{-8}$ at 0.67–0.92 meV.

1 Why these two frequencies

- **The law.** The framework's galactic law is $g_N = g^2/(g + a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$. The value $\kappa = \frac{1}{2}$ is fitted, not derived. With $\kappa = \frac{1}{2}$, $\rho_{\rm DE}c^2 = 4a_0^2/G$.
- **The mass.** The only dimensionally available quantum mass is $m = (\rho_{\rm DE}\hbar^3/c^3)^{1/4}$. We scanned every integer-exponent combination of $(a_0, G, c, \hbar, \rho_{\rm DE})$, and no other combination gives a visible-band mass [1].
- **Two footings.** The two values of a_0 quoted for this programme (9.3603×10^{-11} and $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$) give

$$m = 2.240 \text{ meV } (\nu = 541.7 \text{ GHz},\ \lambda = 0.553 \text{ mm}), \qquad m = 2.463 \text{ meV } (\nu = 595.5 \text{ GHz},\ \lambda = 0.503 \text{ mm}).$$

- **Line width.** A field with the galactic velocity dispersion $\sigma \simeq 230\,\mathrm{km\,s^{-1}}$ gives a line width $\Delta\nu = \nu\sigma^2/c^2 \simeq 0.3$ MHz.
- **The search space.** The search is two narrow windows, not a mass scan.

2 Signal and reach

- **Signal power.** A flat conducting plate of area A in a kinetically mixed vector field of density ρ emits power [2]

$$P = \tfrac{2}{3}\,\varepsilon^2\rho c^3 A,$$

normal to its surface. Here $\frac{2}{3}$ averages over random polarization.
- **Density used.** We take the framework's local cold density, $\rho = 0.01\,M_\odot\,\mathrm{pc^{-3}}$ ($\rho c^2 = 6.1 \times 10^{-5}\,\mathrm{J\,m^{-3}}$). This assumes the hypothesis under test: that the cold component *is* the converting field.
- **Example rate.** At $\varepsilon = 10^{-12}$ and $A = 10\,\mathrm{m}^2$, $P = 1.2 \times 10^{-19}$ W, or about 3×10^7 photons per day.
- **Noise.** The radiometer equation gives $\sigma_P = k_B T_{\rm sys}\sqrt{\Delta\nu/t}$.

plate	receiver	1 h	1 day	30 days
$1\,\mathrm{m}^2$	warm, 1000 K	7×10^{-12}	3×10^{-12}	1.4×10^{-12}
$1\,\mathrm{m}^2$	SIS, 100 K	2×10^{-12}	1.0×10^{-12}	4.5×10^{-13}
$10\,\mathrm{m}^2$	SIS, 100 K	7×10^{-13}	3×10^{-13}	1.4×10^{-13}

5σ reach in ε (lane CFG797 [1]). Receiver temperatures are provisional.

3 Apparatus

- **Plate and mirror.** Use a flat or spherical polished aluminium plate, flat to better than about $\lambda/20 \approx 25\,\mu$m, and a parabolic mirror focusing onto the feed horn. Existing millimetre-wave dish experiments use this geometry [3].
- **Receiver.**
  - An SIS heterodyne mixer from radio-astronomy band-8/9 technology covers both lines with one LO retune.
  - The backend is a spectrometer with $\lesssim 30\,$kHz channels.
  - A room-temperature Schottky receiver suffices for the converter-window test.
- **Path.** The 557 GHz water line lies between the two targets. Purge the optical path with dry nitrogen, or evacuate it.
- **Shielding and checks.**
  - Use an absorber-lined enclosure and a reflector-in/out chop.
  - Inject a calibration tone at each target frequency.
  - A real dark-field line does not move when the plate is rotated about its normal. It does change with the plate's area and orientation.
- **Blinding.** Fix both frequency windows in advance (this note). Offset the channel labels until the analysis is frozen.

4 What each outcome means

- **A line at 541.7 or 595.5 GHz, with the expected width and plate scaling.**
  - The framework's statement that cold energy has no visible channel is falsified [1]. The framework as committed is dead.
  - A dark sector with a light-coupled vector would be discovered.
- **No line, at $\varepsilon \lesssim 10^{-12}$.**
  - The framework's null prediction survives.
  - The only portal that could convert the cold component to light at useful efficiency is closed. That window needs $\varepsilon \approx 4$–6×10^{-9} [1]. Even a one-hour warm-receiver run covers it by more than $500\times$.
- **A line at any other frequency.** It is unrelated to this framework's prediction, but interesting in its own right.

5 Correction in v1.1: the decisive coupling

- **What changed.** Version 1.0 stated that a line at $\varepsilon \gtrsim 10^{-12}$ would falsify the framework. For a kinetically mixed vector field present since the early universe, resonant photon conversion in the μ/y eras distorts the CMB spectrum. At $m \sim 10^{-3}\,$eV this bounds $\varepsilon \lesssim 10^{-13}$ (provisional handbook value; a 2024 reanalysis strengthens it in exactly this window) [4,5].
- **Consequence.** A line at $\varepsilon \geq 10^{-12}$ is already excluded by cosmology for such a field. The decisive laboratory search must reach $\varepsilon \sim 10^{-13}$, where the expected rate is still $\sim 4 \times 10^5$ photons per day per $10\,\mathrm{m}^2$ [1]. Only the $10\,\mathrm{m}^2$ SIS configuration over $\sim$30 days (1.4×10^{-13}) gets there.
- **Why the dish still matters.** The CMB bound is indirect: it assumes the field existed and coupled in the early universe. A dish measures the local field directly. An archival search using the Herschel/HIFI primary mirror at 541.7 GHz (lane CFG799, in progress) gives a direct, cosmology-independent bound of order $\varepsilon \lesssim 2 \times 10^{-11}$ (provisional), already excluding the cold-energy converter window (4–6×10^{-9}) without any cosmological assumption.

- The converter conclusion is unchanged and now closed twice: by cosmology and by direct data.

6 Caveats

- The target mass relies on the identification $\rho_{\rm DE} = 4a_0^2/(Gc^2)$, which is equivalent to the fitted coefficient $\kappa = \frac{1}{2}$.
- The local density and receiver temperatures are provisional.
- Cosmological and stellar bounds on a 2.2–2.5 meV kinetically mixed vector were checked only against the solar relation ($\varepsilon \lesssim 6 \times 10^{-9}$) [1]. CMB-era bounds were not re-audited.
- The comparison with published searches comes from a literature search and is provisional.

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[1] This programme (public repository): ARGON03 haloscope audit (native mass, kill test); lanes CFG723 (no visible channel), CFG795 (converter design space) and CFG797 (this forecast), each with frozen criteria and scripts. [2] D. Horns, J. Jaeckel, A. Lindner, A. Lobanov, J. Redondo, A. Ringwald, JCAP 04 (2013) 016. [3] S. Knirck et al., "First results from a hidden photon dark matter search in the meV sector using a plane-parabolic mirror system", JCAP 11 (2018) 031. [4] A. Caputo, A. J. Millar, C. A. J. O'Hare, E. Vitagliano, PRD 104 (2021) 095029 (dark-photon limits handbook). [5] B. Cyr, J. Chluba et al., MNRAS 535 (2024) 1874 (CMB spectral-distortion constraints). AI assistance: Anthropic's Claude; parallel AI sessions.